

1.2.72

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Question

Evaluate the integral:

$$\int_0^3 \int_0^{3-x} (6 - x - y) dy dx \quad (1)$$

Options:

- (a) 13.5
- (b) 18.0
- (c) 40.5
- (d) 54.0

Step 1: Inner Integral

Evaluate the inner integral:

$$\int_0^{3-x} (6 - x - y) dy \quad (2)$$

Split the integral:

$$= \int_0^{3-x} (6 - x) dy - \int_0^{3-x} y dy \quad (3)$$

$$= (6 - x)(3 - x) - \frac{(3 - x)^2}{2} \quad (4)$$

Step 2: Simplification

$$(6 - x)(3 - x) = 18 - 9x + x^2 \quad (5)$$

$$\frac{(3 - x)^2}{2} = \frac{9 - 6x + x^2}{2} \quad (6)$$

$$\Rightarrow \int_0^{3-x} (6 - x - y) dy = 18 - 9x + x^2 - \frac{9 - 6x + x^2}{2} \quad (7)$$

$$= \frac{27 - 12x + x^2}{2} \quad (8)$$

Step 3: Outer Integral

$$\int_0^3 \frac{27 - 12x + x^2}{2} dx = \frac{1}{2} \int_0^3 (27 - 12x + x^2) dx \quad (9)$$

$$= \frac{1}{2} \left[27x - 6x^2 + \frac{x^3}{3} \right]_0^3 \quad (10)$$

Step 4: Final Calculation

Substitute limits:

$$= \frac{1}{2} (81 - 54 + 9) \quad (11)$$

$$= \frac{1}{2} \times 36 = 18 \quad (12)$$

Region of Integration

In one line: Volume under the plane $z = 6 - x - y$ over a triangular region in the xy -plane.

Numerical Grid Squaring Method

In this method , region is divided into small squares of size:

$$h \times h \quad (13)$$

At each grid point (x_i, y_j) , the function value is:

$$f(x_i, y_j) = 6 - x_i - y_j \quad (14)$$

The double integral is approximated as:

$$\int_0^3 \int_0^{3-x} (6 - x - y) dy dx \approx \sum_i \sum_j (6 - x_i - y_j) h^2 \quad (15)$$

Numerical Grid Squaring Method

- 1 Choose a small step size h
- 2 Loop over x from 0 to 3
- 3 For each x , loop over y from 0 to $3 - x$
- 4 Accumulate:

$$\text{sum} += (6 - x_i - y_j) \quad (16)$$

- 5 Multiply final sum by h^2

Python Code for Grid Squaring Method

```
import numpy as np

h = 0.001 # step size

x = np.arange(0, 3 + h, h)

total = 0

for xi in x:
    y = np.arange(0, 3 - xi + h, h)
    for yj in y:
        total += (6 - xi - yj)

result = total * h * h

print(Approximate Value =, result)
```

Conclusion

For $h = 0.01$

$$\int_0^3 \int_0^{3-x} (6 - x - y) dy dx \approx \sum_i \sum_j (6 - x_i - y_j) h^2 = 18.018831724000133 \quad (17)$$

For a sufficiently small step size h , the numerical result is:

$$\boxed{18 \text{ (approximately)}} \quad (18)$$

The grid squaring method approximates the double integral by summing contributions over small squares. Accuracy improves as the step size h decreases.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')

# --- Top surface (plane) ---
x = np.linspace(0, 3, 100)
y = np.linspace(0, 3, 100)
X, Y = np.meshgrid(x, y)

Z = 6 - X - Y
mask = (X + Y <= 3)
Z = np.where(mask, Z, np.nan)

ax.plot_surface(X, Y, Z, alpha=0.7)
```

Python Code

```
# --- Bottom (triangle) ---
ax.plot_trisurf(
    [0, 3, 0],
    [0, 0, 3],
    [0, 0, 0],
    alpha=0.4
)

# --- Side 1 (x = 0 plane) ---
y_side = np.linspace(0, 3, 50)
z_side = np.linspace(0, 6, 50)
Y_side, Z_side = np.meshgrid(y_side, z_side)
X_side = np.zeros_like(Y_side)

# Keep only valid region
mask_side = (Z_side <= 6 - Y_side)
Z_side = np.where(mask_side, Z_side, np.nan)

ax.plot_surface(X_side, Y_side, Z_side, alpha=0.4)
```

Python Code

```
# --- Side 2 (y = 0 plane) ---
x_side = np.linspace(0, 3, 50)
z_side = np.linspace(0, 6, 50)
X_side, Z_side = np.meshgrid(x_side, z_side)
Y_side = np.zeros_like(X_side)

mask_side = (Z_side <= 6 - X_side)
Z_side = np.where(mask_side, Z_side, np.nan)

ax.plot_surface(X_side, Y_side, Z_side, alpha=0.4)

# Labels
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')

ax.set_title('Solid Volume under  $z = 6 - x - y$ ')

plt.show()
```

Solid Volume under $z = 6 - x - y$

